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Electromagnetic

Assignment 5

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1 Problem 1

سوال ۱-

الف)

$$D_{up}^{\perp} - D_{down}^{\perp} = \sigma_f$$

$$\sigma_f = 0$$

$$D_{up}^{\parallel} - D_{down}^{\parallel} = P_{up}^{\parallel} - P_{down}^{\parallel}$$

$$P_{up}^{\parallel} = 0, P_{down}^{\parallel} = (\epsilon - \epsilon_0) E^{\parallel}$$

$$\epsilon_0 E_0^{\perp} = \epsilon E^{\perp} \rightarrow E^{\perp} = \frac{\epsilon_0}{\epsilon} E_0^{\perp}$$

$$\epsilon E^{\parallel} = \epsilon_0 E_0^{\parallel} + (\epsilon - \epsilon_0) E^{\parallel} \rightarrow E^{\parallel} = E_0^{\parallel}$$

$$\begin{aligned} \vec{E} &= (\vec{E}_0 \cdot \hat{n}) \hat{n} + \frac{\epsilon_0}{\epsilon} |\vec{E}_0 \times \hat{n}| \hat{x} \\ &= E_0 \cos \theta \hat{y} + \frac{\epsilon_0 E_0 \sin \theta}{\epsilon} \hat{x} \end{aligned}$$

ب)

$$\vec{P} = (\epsilon - \epsilon_0) \vec{E}$$

$$P_b = -\nabla \cdot \vec{P} \quad \sigma_b = \vec{P} \cdot \hat{n}$$

$$\sigma_b = (\epsilon - \epsilon_0) E_0 \cos \theta$$

$$P_b = 0$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q$$

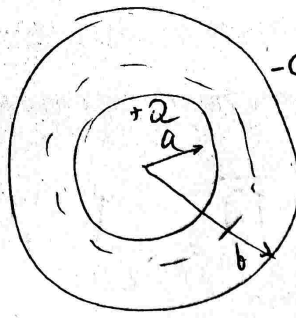
$$Q = (\sigma_b + \sigma_f) \times \pi R^2$$

$$\rightarrow \oint \vec{E} \cdot d\vec{a} = (\epsilon_r - 1) \epsilon_0 \cos\theta \times \pi R^2$$

$$\begin{aligned} \int \vec{D} \cdot d\vec{l} &= (D_{up}^{II} - D_{down}^{II}) L \\ &= -(\epsilon - \epsilon_0) E L \frac{\epsilon_0}{\epsilon} E \sin\theta \end{aligned}$$

2 Problem 2

سوال 2-



$$\oint \vec{D} \cdot d\vec{a} = Q_f = Q$$

به دلیل تقارن $\vec{D}$ تقارن کروی دارد:

$$D 4\pi r^2 = Q \rightarrow \vec{D} = \frac{Q}{4\pi r^2} \hat{r}$$

$$\vec{E} = \frac{\vec{D}}{\epsilon} = \frac{Qa}{4\pi\epsilon_0 r^3} \hat{r}$$

$$V_a - V_b = - \int_a^b \frac{Qa}{4\pi\epsilon_0 r^3} dr = \frac{Qa}{8\pi\epsilon_0} \left(\frac{1}{b^2} - \frac{1}{a^2} \right)$$

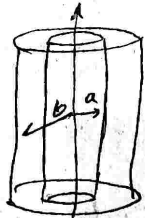
$$\rightarrow C = \frac{8\pi\epsilon_0 ab^2}{b^2 - a^2}$$

$$W = \int_a^b \frac{1}{2} \epsilon E^2 4\pi r^2 dr$$

$$= \frac{Q^2 a}{8\pi\epsilon_0} \int_a^b \frac{1}{r^3} dr = \frac{Q^2 (b^2 - a^2)}{16\pi\epsilon_0 ab^2}$$

$$W = \frac{1}{2} C V^2 = \frac{1}{2} \epsilon \frac{Q^2}{C} = \frac{Q^2 (b^2 - a^2)}{16\pi\epsilon_0 ab^2}$$

3 Problem 3



$$\oint \vec{D} \cdot d\vec{a} = Q_f$$

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$$D \times 2\pi r = k\lambda \rightarrow \vec{D} = \frac{\lambda}{2\pi r} \hat{r}$$

$$\vec{E} = \begin{cases} \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} & r < a \\ \frac{\lambda a}{2\pi\epsilon_0 r^2} \hat{r} & a < r < b \\ \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} & r > b \end{cases}$$

$$\vec{P} = \begin{cases} 0 & r < a \quad r > b \\ \frac{\lambda a}{2\pi a} (r/a - 1) \hat{r} & a < r < b \end{cases}$$

1-

$$\begin{aligned} \nabla \cdot \vec{P} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 P_r) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{\lambda a}{2\pi} (r/a - 1) \right) \\ &= \frac{\lambda}{2\pi r^2} \end{aligned}$$

$$-(\vec{P} \cdot \hat{r})|_{r=a} = \sigma_a = 0$$

$$(\vec{P} \cdot \hat{r})|_{r=b} = \sigma_b = \frac{\lambda a}{2\pi b^2} \left(\frac{b}{a} - 1 \right)$$

4 Problem 4

سوال 4

$$D = \epsilon E = \epsilon_0 E + P$$

$$E = \frac{m}{\epsilon - \epsilon_0} R^2 \hat{r}$$

$$D = \frac{m}{1 - \frac{1}{\epsilon_r}} R^2 \hat{r}$$

$$\int D \cdot da = Q_f$$

$$\frac{m}{1 - \frac{1}{\epsilon_r}} 4\pi R^4 = Q_f(R)$$

$$Q_f(a) = Q = \frac{m}{1 - \frac{1}{\epsilon_r}} 4\pi a^4$$

$$\rightarrow Q_f(R) = Q \frac{R^4}{a^4}$$

$$\int E \cdot da = \frac{Q}{\epsilon_0} \rightarrow \vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \text{ rad } \hat{r}$$